

Slide 1: Rover Design Options for Lunar South Pole Exploration

Context: After probable ice-rich zones and safe landing regions are identified, the rover design must be selected based on illumination availability, thermal survival requirements, mission duration, and feasibility.

Purpose: To compare two possible rover design options and select the most suitable one for the final mission concept.

Rover Design Option	Target Operating Region	Key Benefits	Major Challenges
Solar-Optimized Rover	Sunlit ridges, crater rims, PSR boundaries, and partially illuminated zones	Lower system complexity, lighter power architecture, simpler deployment, suitable for illumination-aware exploration	Strong dependence on sunlight, limited operation inside PSRs, high heater power demand, reduced survival margin in extreme cold
RTG-Based Rover	Deep PSRs, cold traps, low-illumination ice-rich zones, and extended-duration exploration	Sunlight-independent operation, continuous power availability, radioisotope decay heat supports thermal survival, longer mission life	Higher mass, higher cost, nuclear safety requirements, regulatory approval complexity

Selection Logic: Solar-optimized rover is preferred if the mission focuses on illuminated ridges, crater rims, and PSR boundary regions. RTG-based rover is preferred if the mission requires long-duration exploration inside shadowed ice-rich regions.

Key Takeaway: These are two candidate rover designs. The final rover design will be selected based on the target region's illumination, thermal conditions, and mission feasibility.